RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies

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Abstract

*Memory is critical for long-horizon and history-dependent robotic manipulation. Such tasks often involve counting repeated actions or manipulating objects that become temporarily occluded. Recent vision-language-action (VLA) models have begun to incorporate memory mechanisms; however, their evaluations remain confined to narrow, non-standardized settings. This limits their systematic understanding, comparison, and progress measurement. To address these challenges, we introduce **RoboMME**: a large-scale standardized benchmark for evaluating and advancing VLA models in long-horizon, history-dependent scenarios. Our benchmark comprises 16 manipulation tasks constructed under a carefully designed taxonomy that evaluates temporal, spatial, object, and procedural memory. We further develop a suite of 14 memory-augmented VLA variants built on the $\pi_{0.5}$ backbone to systematically explore different memory representations across multiple integration strategies. We show that the effectiveness of memory representations is highly task-dependent, with each design offering distinct advantages and limitations across different tasks. Videos and code can be found at `https://anonymtest1.github.io/`. This paper has been accepted at ICML 2026 as spotlight.*

1. Introduction

Open-world robotic manipulation often requires reasoning over history and recalling information from past interactions. For instance, a household robot may be asked to return books to their original positions on a shelf, or fold laundry after observing a human demonstration. In such scenarios, relying solely on immediate perception for action is insufficient. Effective execution depends on retrieving relevant information across time, which we broadly refer to as *memory*.

Prior work incorporates memory into robotic manipulation policies through three main representations: (1) *symbolic memory*, which summarizes history using non-differentiable abstractions, such as point-tracking trajectories [10] and language-based subgoals [31]; (2) *perceptual memory*, which represents history as a group of visual features, including multi-frame visual tokens [19] and memory banks [14, 28]; and (3) *recurrent memory*, which compresses contextual features into fixed-size latent states through recurrent models [23, 40]. While demonstrating the importance of memory, these methods rely on different policy backbones and inconsistent evaluation protocols, making it unclear which memory designs generalize across tasks. Progress is further limited by the absence of benchmarks that capture diverse and challenging memory requirements. MemoryBench [14] is the first benchmark to explicitly evaluate spatial memory, but it contains only three near-solved tasks. MIKASA-Robo [11] introduces several history-dependent tasks, yet they remain short-horizon and lack sufficient high-quality demonstrations for effective vision-language-action (VLA) imitation learning. Consequently, existing benchmarks neither capture realistic memory demands nor provide a standardized testbed for systematically evaluating memory-augmented manipulation policies.

To address these limitations, we present **RoboMME**: a unified large-scale **Robo**tic simulation benchmark designed for **M**emory-augmented **M**anipulation **E**valuation. Drawing inspiration from cognitive theories of human memory [1], RoboMME categorizes memory into four cognitive dimensions: (1) *temporal memory* for event accumulation and ordering; (2) *spatial memory* for tracking object locations under occlusion and scene change; (3) *object memory* for resolving referential identity across time; and (4) *procedural memory* for reproducing previously demonstrated motion patterns. These memory types define four corresponding task suites, `Counting`, `Permanence`, `Reference`, and `Imitation`, respectively, each composed of four carefully designed tasks. In total, RoboMME contains 16 diverse long-horizon tasks with 1,600 demonstrations.

Building on RoboMME, we develop a family of 14 memory-augmented VLA models based on the $\pi_{0.5}$ backbone [4] to systematically study how different memory representations influence manipulation performance. Symbolic

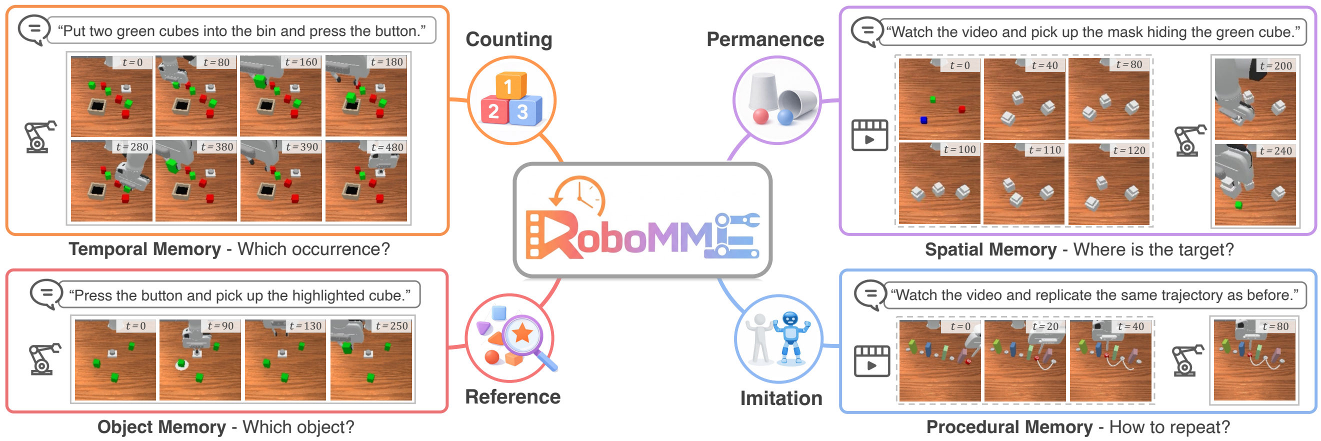

Figure 1. **RoboMME** is a large-scale robotic benchmark for evaluating memory-augmented manipulation, comprising four task suites that emphasize distinct memory demands. (1) The `Counting` suite targets *temporal memory*, requiring robots to accumulate and reason over past events, e.g., counting placed green cubes and stopping correctly (top-left). (2) The `Permanence` suite focuses on *spatial memory*, requiring tracking of object locations under occlusion and environmental changes, e.g., resolving the correct mask after green and blue cubes swap positions (top-right). (3) The `Reference` suite evaluates *object memory*, requiring identification under varied referential cues, e.g., recalling a briefly highlighted cube during a button press (bottom-left). (4) The `Imitation` suite assesses *procedural memory*, measuring the ability to reproduce previously demonstrated behaviors, e.g., replicating a demonstrated trajectory (bottom-right).

memory is implemented as language subgoals concatenated with task instructions, explicitly encoding history in natural language without modifying the backbone. Differentiable neural representations, including perceptual and recurrent memory, are integrated through three mechanisms: (1) *memory-as-context*, which appends memory embeddings to the inputs for joint processing; (2) *memory-as-modulator*, which conditions the action expert via adaptive LayerNorm [27] to modulate intermediate activations; and (3) *memory-as-expert*, which adds a dedicated memory expert that interacts with the action expert through block-wise causal attention [3].

Interestingly, our experiments show that no single memory representation or integration strategy consistently performs well across all tasks, indicating that conclusions from prior work may not generalize. Instead, effectiveness is highly task-dependent: symbolic memory excels at counting and short-horizon reasoning, while perceptual memory is critical for time-sensitive and motion-centric behaviors. Among all variants, perceptual memory combined with the memory-as-modulator design achieves the best balance between performance and computational efficiency. Together, these results establish the first comprehensive framework for understanding memory in manipulation, and represent a step toward reliable, long-horizon, history-dependent robotic generalist policies.

2. Related Work

Memory-based Robotic Manipulation Benchmarks. Most prior robotic manipulation benchmarks [21, 22, 25, 41] implicitly involve temporal memory but rarely require it: policies conditioned only on the current observation can already achieve high success rates [4, 13, 36]. This suggests that success often relies on local perception rather than true history-based reasoning. For example, object states are continually updated during online execution, allowing the agent to infer the next subtask without relying on past observations. MemoryBench [14] is the first attempt to introduce explicit memory requirements, focusing on spatial location recall, but its tasks remain simple and nearly solved. MIKASA-Robo [11], originally designed for reinforcement learning, includes some memory-related tasks, yet these tasks remain short in horizon, are limited in long-term dependencies, and lack sufficient high-quality demonstrations for effective imitation learning. Recent memory-based policies [28, 31] are typically evaluated on narrow, self-designed tasks, hindering systematic comparison and understanding. In contrast, RoboMME provides a challenging testbed with *explicit* history dependence, with diverse memory types and sufficient demonstrations.

Memory-based Robotic Manipulation Models. Existing approaches can be broadly categorized according to their memory representations. (1) *Symbolic memory* relies on non-differentiable abstractions derived from interaction history or external modules. For example, HistRISE [10] tracks object-centric 3D points as symbolic states, UniVLA [5] incorporates temporal context by adding past actions into input prompts. More recent methods, such as MemER [31] and Gemini-Robotics-1.5 [33], generate language-based subgoals with large vision-language models, offloading long-term memory reasoning to agentic pipelines, but introduce costly modular inference and preclude end-to-end optimization. (2) *Perceptual memory* represents history as differentiable visual or multimodal features. ContextVLA [19] appends past visual tokens directly to the transformer input as raw contextual tokens. In contrast, memory-bank approaches encode visual features together with auxiliary signals, such as task instructions [20, 28] or action heatmaps [14], and cache these multimodal embeddings for subsequent retrieval. (3) *Recurrent memory* compresses history into fixed-size latent states through iterative updates. Early work, such as BC-RNN [24], models temporal dependencies using recurrent neural networks. More recent approaches, including MITL [40] and RoboMamba [23], adopt Mamba-style state-space models [15] to better capture long-horizon dependencies. To evaluate differnet memory designs, we construct a family of memory-augmented VLA models on the same backbone and evaluate diverse memory representations and integration strategies under a controlled, standardized setting.

3. RoboMME Benchmark

The aim of RoboMME is to rigorously evaluate history-dependent behavior in robotic manipulation. All tasks are intentionally constructed to be non-Markovian, requiring models to reason over past observations that are no longer visible at the current step. Memory is essential for these tasks because identical observations can arise from different histories yet require different actions.

3.1. Cognitively-Motivated Task Design

To systematically evaluate history-dependent, long-horizon manipulation, we ground our task design in established cognitive models of human memory. Classical theories [1] divide long-term memory into *procedural* and *declarative* types, with declarative memory further divided into *episodic* and *semantic* forms. In particular, episodic memory supports recall of events and experiences, including temporal order, spatial context, and object identity [2, 8, 17], whereas procedural memory encodes motor skills acquired through practice [30]. Both are essential and relevant for memory-augmented manipulation, motivating the four memory types defined below.

- **Temporal memory**: Tracks event counts, sequence ordering, and transition conditions across steps, e.g., determining when to proceed to the next subtask.
- **Spatial memory**: Maintains object locations and spatial relationships, especially when current visual information becomes unreliable due to occlusion or scene changes.
- **Object memory**: Preserves referential consistency over time, enabling reliable object identification from visual, language, or action cues.
- **Procedural memory**: Reproduces previously demonstrated motion patterns or manipulation behaviors.

These four memory types correspond to the cognitive dimensions of when (temporal), where (spatial), what (object), and how (procedural). RoboMME is organized around these dimensions into four task suites, `Counting`, `Permanence`, `Reference`, `Imitation`, each emphasizing a primary memory type for controlled evaluation. Fig. 1 illustrates an overview of RoboMME. Specifically, the `Counting` suite targets *temporal memory* by requiring agents to repeat actions a specified number of times, including pick-and-place, linear swinging, and time-critical tasks. The `Permanence` suite focuses on *spatial memory*, evaluating object location tracking from pre-recorded videos or during concurrent manipulation. The `Reference` suite evaluates *object memory* through persistent identity resolution under visual, action-based, and language-based referential cues. The `Imitation` suite targets *procedural memory* by requiring agents to reproduce demonstrated motion patterns, such as pushing, inserting, and sequential linear or circular motions. Tab. 1 summarizes the tasks, with detailed descriptions in Sec. E.

3.2. Benchmark Construction

We build on the ManiSkill simulator [26] using a tabletop environment with a 7-DOF Franka Panda arm. Training episodes are generated by replaying predefined keyframe waypoints and are recorded as dense trajectories.

Observations and Actions. At each timestep, the robot receives multi-view RGB observations from front- and wrist-mounted cameras (both 256×256), along with proprioceptive states including joint positions, end-effector (EEF) pose, and gripper state. The action space is defined in either absolute joint space or absolute EEF space: joint-space actions are 8D (7 joints + gripper), EEF-space actions are 7D (3D position, Euler orientation, and gripper). Tasks in the `Imitation` suite and those prefixed with "Video" use video-based observations (a sequence of historical frames with paired proprioception) at the initial step, whereas all other tasks use image-based observations (a single current frame with paired proprioception). During execution, all tasks provide image-based observations at every timestep.

Benchmark Comparison. Tab. 2 provides a detailed comparison with prior benchmarks. Most existing benchmarks [16, 25, 38] are designed to emphasize long-horizon planning and task complexity. While these tasks may implicitly require temporal reasoning, they do not explicitly enforce other types of memory, as decisions can typically be made from immediate observations alone. In contrast, RoboMME introduces greater environmental complexity and video-conditioned tasks, and is the first to systematically cover four types of memory: temporal, spatial, object, and procedural.

Data Curation. To enhance behavioral diversity, particu-

Task Name	Memory Type	Avg. #Steps	Task Challenge	Brief Description
`Task Suite: Counting`				
PickXTimes	T	538	count	Pick and place a cube of a given color for a specified number of repetitions.
BinFill	T	604	count	Place a specified number of cubes of a given color into the bin as the cubes appear over time.
SwingXTimes	T	435	count, swing-motion	Swing a cube back and forth between two targets for a specified number of cycles.
StopCube	T	317	count, time-critical	Press a button *exactly* when a moving cube reaches the target at a specified occurrence.
`Task Suite: Permanence`				
VideoUnmask	S	217	occlusion	Given a video in which all cubes are masked, uncover the cube of a specified color.
ButtonUnmask	S	267	occlusion	Press the button, during which all cubes are masked, then uncover the cube of a specified color.
VideoUnmaskSwap	S	348	occlusion, tracking	Given a video in which all cubes are masked and containers dynamically swap positions, uncover the cube of a specified color.
ButtonUnmaskSwap	S	400	occlusion, tracking	Press the button, during which all cubes are masked and containers dynamically swap positions, then uncover the cube of a specified color.
`Task Suite: Reference`				
PickHighlight	O	346	visual-referential	Pick up all cubes that were visually highlighted in a short time during interaction.
VideoRepick	O + T	687	action-referential, tracking, count	Given a video showing a cube being manipulated and relocated, pick up the same cube for a specified number of repetitions.
VideoPlaceButton	O + T	974	language-referential, long-video, tracking	Given a video with interleaved cube placement and button pressing, place the cube on the target specified by a language-described temporal reference (e.g., *the target after pressing the button*).
VideoPlaceOrder	O + T	1134	language-referential, long-video, tracking	Given a video showing cube placement across multiple targets, place the cube on the target specified by a language-described ordinal reference (e.g., *the second target*).
`Task Suite: Imitation`				
MoveCube	P	394	contact-mode, tool-use	Given a video showing cube transport, replicate the same demonstrated manipulation strategy (e.g., *pick-and-place, pushing with the gripper, or hooking with a stick*).
InsertPeg	P + O	479	precise-motion	Given a video showing peg insertion, grasp the same peg at the same end and insert it into a box following the same demonstrated direction.
PatternLock	P	208	linear-motion	Given a video showing a linear moving pattern, reproduce the same trajectory on the targets.
RouteStick	P	370	circular-motion	Given a video showing a circular routing pattern, reproduce the same trajectory around sticks.

Table 1. **Task Summary**. Overview of all tasks, their corresponding memory types, average total timesteps, and key challenges.

Dataset	Environment Complexity			Condition		Annotation		Memory Type	Total #Tasks	Total #Demos	Avg. #Steps
	Non-Markov	Partial Obs.	Dynamic	Vid.	Lang.	Subgoal	Keyframe				
RLBench18 [18, 29]	✗	✗	✗	✗	✓	✗	✓	T	18	1,800	137
CALVIN [25]	✗	✗	✗	✗	✓	✓	✓	T	34	1,000	584
LIBERO [22]	✗	✗	✗	✗	✓	✗	✗	T	130	6,500	162
VLABench [38]	✗	✗	✗	✗	✓	✗	✗	T	100	1,600	115
RoboCerebra [16]	✗	✗	✓	✗	✓	✓	✗	T	100	1,000	2,972
MemoryBench [14]	✓	✗	✗	✗	✓	✗	✓	T+S	3	300	312
MIKASA-robo (VLA) [11]	✓	✓	✓	✗	✗	✗	✗	T+S+O	12	1,250	72
RoboMME (Ours)	✓	✓	✓	✓	✓	✓	✓	**T+S+O+P**	**16**	**1,600**	**481**

Table 2. **Benchmark Comparison.** The upper section lists general-purpose benchmarks and the lower memory-based benchmarks. Memory types include temporal (T), spatial (S), object (O), and procedural (P). Environment complexity captures non-Markovian structure, partial observability, and dynamic scene change. Condition denotes initial inputs (Vid. for video observations; Lang. for language instructions). Annotations indicate the availability of language-based subgoals or keyframe timesteps for auxiliary supervision.

larly failure recovery, which is important for imitation learning [12], we inject 5% noise to keyframe waypoints and then recovering to the normal trajectory. Tasks are further stratified into easy, medium, and hard levels based on scene clutter, horizon length, and environmental dynamics. To ensure data quality, we filter out built-in planner failures for training and evaluation environment seeds. The final dataset spans 16 tasks with 100 episodes each, yielding 770k timesteps. Individual episodes range from a few hundred to over one thousand steps, reflecting long-horizon, history-dependent behaviors.

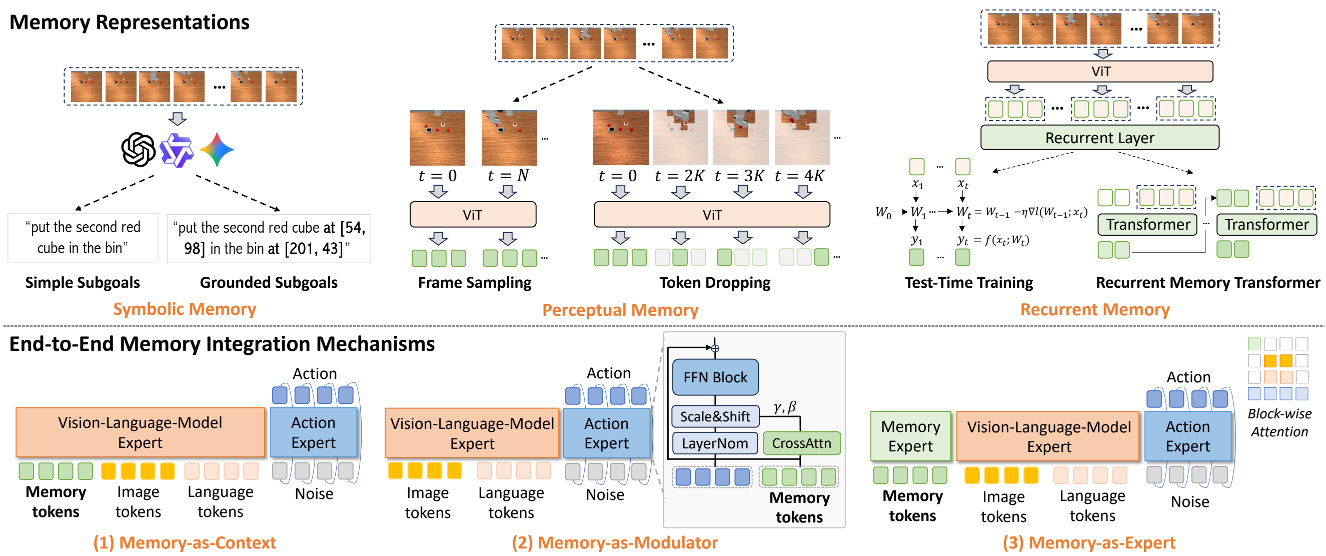

Figure 2. **Framework of MME-VLA Suite.** The top part illustrates three memory representations, each with two instantiations: (1) *Symbolic Memory* summarizes past interactions as high-level abstractions via language-based subgoals, optionally grounded to image pixels; (2) *Perceptual Memory* encodes history as raw visual tokens, using either token dropping to remove redundancy or uniform frame sampling to preserve essential context; (3) *Recurrent Memory* compresses history into fixed-size latent states through test-time training or recurrent memory transformers. The bottom part shows three integration strategies consistent with $\pi_{0.5}$: (1) *Memory-as-Context* concatenates memory tokens with observation tokens (e.g., images, task instructions, and proprioception); (2) *Memory-as-Modulator* applies adaptive LayerNorm to modulate the action expert, where action features cross-attend to memory tokens via multi-head attention; (3) *Memory-as-Expert* adds a separate memory expert that interacts with other experts through block-wise causal attention.

4. Memory-Augmented Manipulation Policies

Building on RoboMME, we construct a family of memory-augmented vision-language-action (VLA) models based on the $\pi_{0.5}$ backbone, collectively termed the **MME-VLA suite**. We systematically compare different memory representations and integration mechanisms under controlled settings, as illustrated in Fig. 2. More detailed model formulations are provided in Sec. A.

4.1. Memory Representations

We study three representations: *symbolic*, *perceptual*, and *recurrent*. Symbolic memory uses discrete language tokens, while the latter two provide differentiable neural features.

Symbolic Memory is represented as interpretable language subgoals, as correct subgoals imply effective history summarization. At each step, we leverage an auxiliary vision-language model (VLM) that conditions on the current image and prior subgoals to generate the next subgoal. We consider *simple* instructions and *grounded* variants that additionally include image coordinates (e.g., *pick up the green cube* vs. *pick up the green cube at [63, 152]*), with grounding being particularly beneficial for spatial reasoning. Notably, symbolic memory approaches typically require additional subgoal annotations to train the VLM.

Perceptual Memory is represented as a sequence of visual tokens selected from past images and extracted by the $\pi_{0.5}$ vision encoder. To select informative tokens, we adopt two strategies: (1) token dropping [37], which removes temporally redundant image patches based on RGB differences; (2) uniform frame sampling [9], which evenly downsamples the sequence and concatenates tokens from the sampled frames. In practice, we found that visual tokens alone suffice to encode history without proprioceptive states.

Recurrent Memory compresses the visual token sequence into fixed-size latent states through recurrence. We adopt two models: (1) Test-Time Training (TTT) [32, 39], which updates fast weights online via a self-supervised loss and applies them to generate output features; (2) Recurrent Memory Transformers (RMT) [6, 7], which process input sequence into segments and recurrently update a set of learnable memory slots for each segment using transformer [34].

4.2. Memory Integration Mechanisms

Symbolic memory can be naturally integrated into $\pi_{0.5}$ by concatenating subgoals with task instructions. In contrast, perceptual and recurrent memory produce neural *memory tokens* that require more architectural modifications. We therefore study three integration mechanisms.

Memory-as-Context. Memory tokens are concatenated with the original inputs and jointly processed by the VLM expert, directly influencing VLM features.

Memory-as-Modulator. We adopt adaptive LayerNorm

(AdaLN) [27] to condition the action expert on external memory. Before entering the feed-forward block in each layer, the action features cross-attend to memory tokens via multi-head attention to extract memory-aware representations. These representations are then projected into scale and shift parameters that modulate the normalized action features through AdaLN.

Memory-as-Expert. We introduce an additional lightweight memory expert that processes memory tokens through a dedicated pathway. The three experts interact via a specified blockwise causal attention scheme: the action expert attends to both the VLM and memory experts, while the VLM and memory experts do not attend to each other. This separation limits interference, preserves the original VLM behavior, and allocates specialized capacity for memory processing.

5. Experiments

In this section, we systematically evaluate all MME-VLA variants under controlled settings.

5.1. Experiment Setup

We evaluate a total of 14 VLA policies in the MME-VLA suite, along with 4 prior methods, for a systematic comparison. Implementation details are provided in Sec. B.

MME-VLA Suite. For symbolic memory, we fine-tune two $\pi_{0.5}$ variants, SIMPLESG and GROUNDSG, which condition on simple or grounded language subgoals. Subgoals are generated by either Gemini-2.5-Pro (`Gemini`), a Qwen3-VL-4B model (`QwenVL`), or the simulator ground truth (`Oracle`). `QwenVL` is fine-tuned on subgoal annotations in our dataset to predict the next subgoal given the current image and the accumulated subgoal history, whereas `Gemini` relies solely on prompt engineering without task-specific fine-tuning. For perceptual memory, we use token dropping (TOKENDROP) or frame sampling (FRAMESAMP), each combined with three integration mechanisms: memory-as-context (`Context`), memory-as-modulator (`Modul`), and memory-as-expert (`Expert`), resulting in six additional $\pi_{0.5}$ variants. For recurrent memory, we adopt TTT or RMT with the same integration strategies, also producing six additional policies. Unless otherwise specified, we denote each variant as "METHOD+`Integration/VLM`", e.g., FRAMESAMP+`Modul` or SIMPLESG+`QwenVL`.

Evaluation Protocols. For fair comparison across our MME-VLA models, we fix a memory budget of 512 tokens, matching the number of image tokens in the current observation, e.g., both the learnable tokens in RMT and the sampled tokens in TOKENDROP are capped at this limit. Evaluation uses 50 episodes per task (800 total) with environment seeds distinct from training. For every method, both MME-VLA variants and prior methods, we adopt a unified multi-task training protocol: a single policy is jointly trained across all 16 tasks, with a maximum of 1,300 steps per episode. Results are averaged over the last three checkpoints and three random seeds (nine runs total).

Prior Methods. We compare against three methods: (1) $\pi_{0.5}$, which conditions only on the current observation without any memory; (2) SAM2Act+ [14], which adopts SAM2 as a backbone to provide perceptual memory through a memory bank and predicts discrete keyframe actions that are executed by an external motion planner; (3) MemER [31], which employs a VLM to infer symbolic subgoals from accumulated keyframe images and execute them with a VLA. Unlike our symbolic variants, which condition only on the current image and subgoal history at each step, MemER selects keyframes from the most recent N images during execution and predicts symbolic subgoals using all previously selected keyframe images. It can therefore be viewed as a hybrid of perceptual and symbolic memory. When reproducing MemER, we follow their original prompts to fine-tune an additional Qwen3-VL-4B model on grounded subgoal and keyframe annotations, and execute the predicted subgoals with the GROUNDSG policy. All prior methods are also trained as a single multi-task policy over the 16 tasks, and evaluated over three runs. We leave the inclusion of additional methods to future work.

5.2. Main Results and Analysis

The main results are summarized in Tab. 3; and complete results are provided in Sec. C. We analyze these results by addressing several key research questions below.

Q1: Which memory representations and integration mechanisms yield the strongest performance? Across all MME-VLA variants, **perceptual memory methods perform best**. In particular, FRAMESAMP+`Modul` achieves the highest overall success rate (44.51%), and most perceptual memory variants outperform the best symbolic and recurrent counterparts. Within perceptual memory, TOKENDROP is less effective than FRAMESAMP, likely because aggressive token pruning removes global spatial context. This loss can degrade performance on tasks such as *StopCube*, which require awareness of object distance and benefit from more holistic views. Across integration mechanisms, **memory-as-modulator performs best** for both TOKENDROP and FRAMESAMP. We attribute this to its lightweight, feature-wise conditioning, which largely preserves the original $\pi_{0.5}$ architecture, whereas other modifications may disrupt pretrained representations of $\pi_{0.5}$.

We also observe that recurrent methods perform the worst, likely because fine-tuning $\pi_{0.5}$ with shallow recurrent layers leads to unstable training. This suggests that effective recurrence requires deeper architectural integration and stronger recurrence-oriented pretraining, as explored in recent computer vision works [35, 39]. In contrast, our symbolic variants achieve up to 32.70% success, with GROUNDSG consistently outperforming SIMPLESG on most tasks using

Method		Counting				Permanence				Reference				Imitation				AVG
		Bin Fill	Pick Xtimes	Swing Xtimes	Stop Cube	Video Umsk	Button Umsk	Video UmskS	Button UmskS	Pick HighL	Video Repick	Video PlcBtn	Video PlcOrd	Move Cube	Insert Peg	Pattern Lock	Route Stick	
Human Performance		96.00	100.0	80.00	78.00	90.00	92.00	92.00	90.00	92.00	92.00	98.00	90.00	90.00	98.00	84.00	86.00	90.50
MME-VLA w/ Symbolic Memory																		
SimpleSG	Oracle	**85.78**	99.78	**100.0**	44.67	33.11	22.00	15.56	15.56	44.00	27.78	31.33	26.00	87.33	10.00	95.33	55.11	49.58
GroundSG	Oracle	**85.78**	**100.0**	**100.0**	**49.67**	**98.78**	**95.00**	**99.22**	**80.22**	**83.33**	**97.33**	**100.0**	**100.0**	**87.78**	**15.56**	**97.00**	**55.56**	**84.08**
SimpleSG	Gemini	46.00	63.00	**45.00**	2.00	29.00	9.00	14.00	2.00	**20.00**	15.00	26.00	29.00	61.00	4.00	7.00	0.00	23.25
	QwenVL	**77.56**	**95.33**	5.11	0.44	34.22	19.33	15.33	9.56	17.11	**25.33**	33.33	25.11	**82.00**	**3.78**	**12.67**	**7.78**	29.00
GroundSG	Gemini	26.00	18.00	4.00	**3.00**	36.00	14.00	13.00	0.00	9.00	17.00	12.00	7.00	17.00	0.00	7.00	2.00	11.56
	QwenVL	52.00	92.67	7.33	0.00	**88.67**	**24.00**	**30.67**	**14.00**	15.11	**25.33**	**54.00**	**31.78**	71.56	3.33	6.67	6.00	**32.70**
MME-VLA w/ Perceptual Memory																		
TokenDrop	Context	48.67	85.11	**94.67**	3.11	**33.78**	**31.56**	26.22	16.00	20.67	17.78	31.11	25.33	81.33	4.00	12.67	20.00	34.50
	Modul	34.44	83.56	86.00	5.33	28.22	29.33	**28.44**	**21.33**	21.33	22.00	59.56	**36.00**	62.00	7.11	32.44	51.56	38.04
	Expert	54.22	**87.56**	91.78	4.22	26.67	30.44	18.44	18.89	19.33	20.89	36.44	24.67	**87.56**	2.22	16.22	18.22	34.86
FrameSamp	Context	41.22	72.00	73.67	13.67	26.89	30.22	20.89	15.22	17.67	15.22	30.00	20.89	77.22	1.22	15.22	19.67	30.68
	Modul	39.56	87.33	92.00	**42.00**	32.67	25.11	24.44	18.22	**22.89**	**30.44**	**60.00**	32.00	77.78	**7.56**	**53.56**	**66.67**	**44.51**
	Expert	**57.33**	86.22	**94.67**	28.89	31.78	25.78	22.89	20.22	19.11	23.11	30.00	24.22	83.11	2.00	13.56	17.11	36.25
MME-VLA w/ Recurrent Memory																		
TTT	Context	35.56	62.89	**42.44**	3.33	29.78	**22.89**	18.44	14.44	**20.44**	**13.11**	**34.22**	20.22	32.44	1.11	1.56	3.56	22.28
	Modul	34.22	**65.11**	36.67	2.11	27.22	22.11	**25.22**	14.11	14.56	12.11	32.67	22.33	31.22	1.11	3.56	7.00	21.96
	Expert	34.89	63.78	41.33	4.00	**31.78**	22.44	19.56	**18.00**	12.22	9.56	34.00	22.89	**33.56**	0.89	3.11	5.56	**22.35**
RMT	Context	32.44	56.89	33.56	**5.78**	31.33	10.89	17.33	2.00	14.00	3.78	32.00	29.11	25.78	2.00	**5.56**	**8.89**	19.46
	Modul	33.33	60.78	37.78	4.67	31.11	11.78	17.78	2.44	17.11	4.22	32.00	**31.11**	24.67	**2.21**	3.78	8.00	20.17
	Expert	**35.78**	60.22	36.00	5.56	28.00	17.11	15.78	2.00	11.78	0.22	24.22	22.67	20.00	1.89	4.22	4.89	18.15
Other Methods																		
$\pi_{0.5}$		30.00	42.89	35.56	**6.67**	20.44	22.22	18.67	6.67	11.33	0.44	**31.11**	25.78	26.00	1.56	2.89	4.67	17.93
SAM2Act+ [14]		40.00	76.00	25.33	0.00	27.33	32.00	18.00	**26.67**	17.33	5.33	24.67	20.00	29.33	0.00	0.00	0.00	21.37
MemER [31]		**56.67**	**79.33**	**59.33**	0.00	**81.33**	**72.00**	**38.00**	21.33	**70.67**	**25.33**	30.00	**26.00**	**82.67**	**6.67**	**16.67**	**12.00**	**42.38**

Table 3. **Main Results.** Our MME-VLA suite integrates $\pi_{0.5}$ with three memory representations: (1) symbolic memory, implemented as simple or grounded subgoals (SimpleSG, GroundSG); (2) perceptual memory, using token dropping (TokenDrop) or frame sampling (FrameSamp); and (3) recurrent memory, using test-time training (TTT) or a recurrent memory transformer (RMT). Symbolic subgoals can be generated by Gemini-2.5-Pro (Gemini), a fine-tuned Qwen3-VL-4B (QwenVL), or simulator ground truth (Oracle). Perceptual and recurrent memory are integrated end-to-end via three mechanisms: memory-as-context (Context), memory-as-modulator (Modul), and memory-as-expert (Expert). **Red** marks the best per section; □ marks the overall best for non-oracle models.

QwenVL, highlighting the importance of accurate grounding for manipulation. On simpler tasks such as *PickXTimes*, where grounding is less critical, SimpleSG can outperform GroundSG, as grounding errors from QwenVL degrade performance. Without fine-tuning, prompt-only Gemini suffers from domain shift and often predicts incorrect labels, further reducing performance.

SAM2Act+ performs poorly on average but achieves the best result on *ButtonUnmaskSwap*, as it uses a motion planner to execute discrete actions reliably. In contrast, VLA policies sometimes fail to press buttons correctly. MemER, the strongest compared method, achieves 42.38% success and excels on *ButtonUnmask* and *PickHighlight*, showing the benefit of combining memory representations, where gains arise from maintaining keyframes to preserve long visual histories while using symbolic subgoals.

Q2: Is high-level symbolic reasoning alone sufficient for memory-augmented manipulation? It depends on the specific task. As an upper bound of symbolic memory, GroundSG+Oracle successfully solves many tasks, confirming that language provides strong representational capacity for high-level reasoning. However, performance still degrades on manipulation-intensive tasks such as *StopCube* and *InsertPeg*, where language offers limited guidance and precise visuomotor control becomes the primary bottleneck. The policy also struggles in cluttered scenes, often manipulating wrong objects and causing unintended collisions in *BinFill* and *PickHighlight* despite unambiguous subgoals. We also conducted a human study by reformulating each task as an *online VideoQA* problem, achieving 90.5% overall success but still fail to fully solve the benchmark(Sec. B.5).

Q3: How does the effectiveness of different memory designs depend on task characteristics? No single memory representation consistently dominates across all tasks. Symbolic memory performs well on several Counting tasks, such as *BinFill* and *PickXTimes*. Perceptual memory excels on Imitation tasks, including *PatternLock* and *RouteStick*. However, no method solves any suite entirely, suggesting that cognitive dimensions do not map one-to-one onto memory representations.

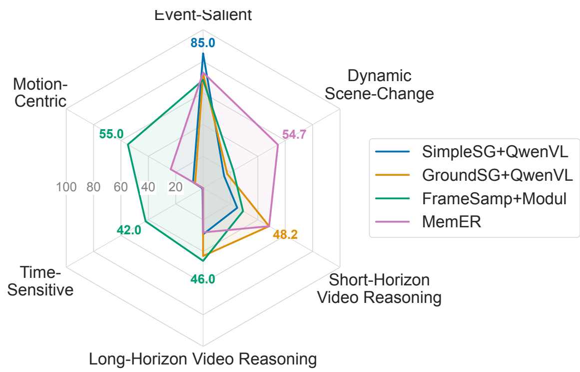

Figure 3. Performance comparison across task characteristics.

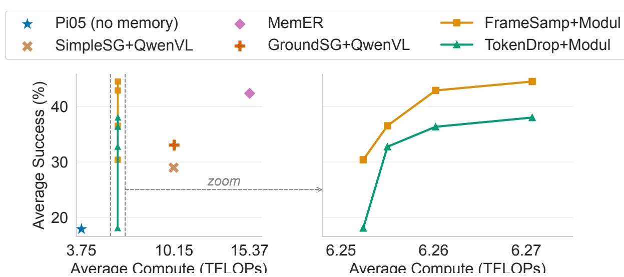

Figure 4. Efficiency-performance comparison among models.

Method	Put Fruits	Track Cube	Repick Block	Draw Pattern	Total
$\pi_{0.5}$	2/10	1/10	1/10	0/10	4/40
GroundSG + QwenVL	**9/10**	3/10	5/10	2/10	19/40
FrameSamp + Modul	6/10	**5/10**	**6/10**	**8/10**	**25/40**

Table 4. Real-world experiment results.

To better analyze the effectiveness of memory representations, we group the 16 tasks by their primary functional requirements: *Motion-Centric*, *Time-Sensitive*, *Short-Horizon Video Reasoning*, *Long-Horizon Video Reasoning*, *Dynamic Scene-Change* and *Event-Salient*. More detailed descriptions are provided in Sec. C.3. As shown in Fig. 3, perceptual memory (FrameSamp + Modul) is most effective for motion-centric, time-sensitive, and long-horizon tasks, where access to extended visual history and low-level motions is critical. In contrast, symbolic memory excels on short-horizon and event-salient tasks by providing clear subgoal guidance. MemER performs best on dynamic scene-change tasks, likely because it retains keyframe images to keep necessary details during online execution. Overall, these results indicate that different memory designs provide complementary strengths, suggesting that combining them synergistically could further improve performance.

Q4: How does memory affect the efficiency-performance balance of memory-augmented policies? Memory improves history dependence but inevitably introduces additional computational cost. To quantify this trade-off, we vary the memory budget (64-512 tokens) and compare efficiency-performance scaling across models. Computational cost is estimated from the TFLOPs of a single forward pass based on the average episode length. As shown in Fig. 4, perceptual memory achieves the most favorable balance. FrameSamp+Modul delivers consistent performance gains as the memory budget increases while incurring only modest additional cost, since most computation arises from processing visual tokens rather than from memory integration itself. In contrast, methods that rely on external VLM inference introduce substantially higher overhead: GroundSG+QwenVL requires roughly 3× the computation of $\pi_{0.5}$, while MemER nearly 5×. In practice, these costs can be reduced through *caching*, e.g., by reusing subgoals with less frequent VLM inference or reusing visual tokens to avoid redundancy.

Q5: Do the trends observed on RoboMME transfer to real-world robotic manipulation? Yes. We evaluate four real-world tasks, *PutFruits*, *TrackCube*, *RepickBlock*, and *DrawPattern*, designed to mirror the *BinFill*, *VideoUnmask(Swap)*, *VideoRepick* and *PatternLock* tasks in simulation. We collect 350 demonstrations and test our strongest symbolic (GroundSG+QwenVL) and perceptual (FrameSamp+Modul) models to assess whether trends can transfer to physical settings. As shown in Table 4, the results exhibit similar patterns. Symbolic memory performs best on the counting task *PutFruits*, whereas perceptual memory excels on motion-centric tasks such as *DrawPattern*. On the remaining two tasks, both approaches achieve comparable performance, suggesting room for further improvement. Additional details are provided in Appendix D.

6. Conclusion and Future Work

This work introduces RoboMME, a unified benchmark for systematically evaluating memory-augmented robotic manipulation across four cognitive dimensions: temporal, spatial, object, and procedural memory. We further develop a family of vision-language-action (VLA) models and conducted controlled comparisons of symbolic, perceptual, and recurrent memory representations with multiple integration mechanisms. Results show that no single design consistently dominates: symbolic memory excels at counting and short-horizon reasoning, while perceptual memory is crucial for motion-centric and time-sensitive behaviors.

Despite its breadth, RoboMME focuses on tabletop manipulation with a fixed set of assets and evaluates a single pretrained backbone ($\pi_{0.5}$), leaving mobile manipulation and alternative architectures, such as memory-banks and other VLA backbones, for future study. Our results further suggest that memory representations are complementary, motivating unified frameworks that integrate multiple forms of memory. We hope RoboMME serves as a foundation and lasting testbed for developing memory-augmented robotic policies.

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